APPLICATION DEVELOPMENT LAB REPORT



Department of Computer Science & Engineering

MALLA REDDY UNIVERSITY, HYDERABAD

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OCR - Based Contract Analysis and Simplification System

Designed and Developed by

1. K. VAISHALI	2211CS010582
2. S. SAI VARSHINI	2211CS010548
3. S. VAMSHI KUMAR	2211CS010543
4. Y. UDAY KIRAN	2211CS010608

Guided by

Mrs. P. SABITHA

Department of Computer Science & Engineering

MALLA REDDY UNIVERSITY, HYDERABAD

2025-2026

MALLA REDDY UNIVERSITY

(Telangana State Private Universities Act No.13 of 2020 and G.O.Ms.No.14, Higher Education (UE) Department)

CERTIFICATE

This is to certify that this is the Application development lab record entitled **"OCR – Based Contract Analysis and Simplification System"**, submitted by **K. VAISHALI(2211CS010582), S. SAI VARSHINI(2211CS010548), S. VAMSHI KUMAR(2211CS010543), Y. UDAY KIRAN(2211CS010608)** B. Tech **IV** year **II** semester, Department of CSE during the year 2025-26. The results embodied in this report have not been submitted to any other university or institute for the award of any degree or diploma.

Internal Guide
MRS. P. SABITHA

DEAN-CSE
Dr. SHAIK MEERAVALI

External Examiner



MALLA REDDY UNIVERSITY

(Telangana State Private Universities Act No.13 of 2020 and G.O.Ms.No.14, Higher Education (UE) Department)

DECLARATION

I declare that this project report titled **"OCR - Based Contract Analysis and Simplification System"** submitted in partial fulfillment of the degree of B. Tech in CSE is a record of original work carried out by me under the supervision of **Mrs. P. SABITHA** and has not formed the basis for the award of any other degree or diploma, in this or any other Institution or University. In keeping with the ethical practice in reporting scientific information, due acknowledgements have been made wherever the findings of others have been cited.

SIGNATURE

K. VAISHALI	2211CS010582
S. SAI VARSHINI	2211CS010548
S. VAMSHI KUMAR	2211CS010543
Y. UDAYKIRAN	2211CS010608

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ABSTRACT

Legal contracts are fundamental documents used across industries including finance, employment, services, real estate, and corporate governance. These documents define rights, responsibilities, payment conditions, liabilities, and dispute resolution mechanisms between parties. However, contracts are typically written in complex legal language containing technical terminology, long sentences, and ambiguous clauses that are difficult for non-legal professionals to interpret.

Manual contract review requires significant time and expertise, increasing operational costs and delaying decision-making processes. Furthermore, individuals without legal knowledge often sign contracts without fully understanding the implications, which may lead to financial or legal risks..

The proposed **OCR - Based Contract Analysis and Simplification System** addresses these challenges by leveraging Natural Language Processing (NLP) techniques to automate contract review. The system extracts key information, identifies important clauses, detects risk indicators, classifies contract sectors, generates concise summaries, and provides simplified explanations in plain English.

The system uses text preprocessing techniques such as tokenization, stop-word removal, and keyword extraction to analyze contract content. It applies rule-based clause detection and risk scoring algorithms to evaluate contract safety. Additionally, a keyword search module allows users to quickly locate specific terms within the document.

This project demonstrates the practical application of NLP in Legal Technology (LegalTech) and provides a foundation for future AI-driven contract intelligence systems.

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CHAPTER 1

INTRODUCTION

1.1 Introduction

In the modern digital era, legal contracts play a crucial role in governing business transactions, employment agreements, service-level commitments, and regulatory compliance. Almost every individual or organization encounters contracts at various stages of life, whether while signing employment offers, rental agreements, insurance policies, or online service terms. Despite their importance, legal contracts are often written using highly complex language, technical legal terminology, and lengthy clauses that are difficult for common users to understand.

The complexity of legal language creates a significant gap between legal professionals and ordinary users. Most individuals lack legal expertise and may unknowingly agree to unfavorable terms due to misunderstanding or lack of awareness. Manual reading and interpretation of contracts is not only time-consuming but also prone to human error, especially when contracts span dozens of pages. This problem becomes more severe for startups, small businesses, and individuals who cannot afford continuous legal consultation.

With advancements in Artificial Intelligence (AI) and Natural Language Processing (NLP), it has become possible to automate the understanding of human language. NLP enables machines to analyze, interpret, and derive meaningful insights from unstructured textual data. When applied to legal documents, NLP can identify important clauses, detect risks, extract obligations, and simplify complex legal language into understandable summaries.

The NLP-Based Automatic Contract Analysis and Simplification System is designed to address these challenges by providing an intelligent, user-friendly platform that analyzes legal contracts automatically. The system uses Optical Character Recognition (OCR) to extract text from uploaded contract documents and applies NLP techniques to process and analyze the extracted content. It highlights key clauses, identifies potential risks, and generates simplified explanations in plain language. contract understanding, and promote informed decision-making. The proposed system not only improves efficiency but also enhances transparency and user confidence while dealing with legal documents.

1.2 Objective of the Project

1. Automate Contract Text Extraction

The system automates the extraction of textual content from scanned or image-based legal contracts using Optical Character Recognition (OCR). By eliminating the need for manual typing or data entry, the platform efficiently converts image documents into machine-readable text, forming the foundation for further automated analysis and reducing processing time.

2. Analyze Contract Content Efficiently

The system processes extracted contract text using rule-based text analysis techniques to identify important clauses, detect predefined legal keywords, and classify the document into relevant sectors such as IT, Finance, Legal, Healthcare, or Construction. This enables structured understanding of contract content without requiring manual legal review.

3. Detect Potential Risk Indicators

The platform evaluates the presence of risk-related terms such as penalties, termination clauses, liabilities, and fines to assess the overall risk level of the contract. This heuristic-based risk detection mechanism helps users quickly identify potentially sensitive or critical clauses within the agreement.

4. Simplify Complex Legal Terminology

The system enhances accessibility by converting complex legal terms into plain and understandable language through lexical simplification. By translating technical legal jargon into simpler explanations, the platform improves readability for non-legal users and supports better comprehension of contractual obligations.

5. Extract and Present Key Contract Points

The solution automatically segments contract text into meaningful clauses and extracts key points for easy reference. This structured presentation enables users to quickly grasp essential information without reviewing the entire document manually.

6. Enable Keyword-Based Search

The platform provides a keyword search functionality that allows users to locate specific terms or clauses within the analyzed contract. This feature improves usability and helps users quickly navigate relevant sections of the document.

7. Generate Structured Analysis Reports

The system generates a comprehensive PDF report containing the contract summary, sector classification, risk level, key points, and simplified explanations. This automated reporting functionality ensures proper documentation and easy sharing of analysis results.

8. Improve Accessibility and Reduce Manual Effort

By integrating OCR, rule-based analysis, and automated reporting within a client-side web application, the system reduces dependency on manual contract review. It provides a user-friendly and efficient solution that enhances transparency and supports informed decision-making.

1.3 Scope of the Project

1. Image-Based Contract Processing

The system supports the analysis of scanned or image-based legal contracts by extracting text using Optical Character Recognition (OCR). It is designed to process documents in common image formats such as JPG and PNG, enabling users to analyze physical or digitally scanned agreements.

2. Rule-Based Contract Analysis

The project focuses on rule-based text processing techniques for contract classification, keyword detection, clause extraction, and risk assessment. The scope includes identifying predefined legal terms and patterns to determine sector classification and potential risk levels within the contract.

3. Legal Term Simplification

The system provides lexical simplification of complex legal terminology into plain language. The scope includes translating commonly used legal terms into user-friendly explanations to enhance readability and understanding for non-legal users.

4. Contract Summary and Key Point Extraction

The project extracts significant clauses and presents them in a structured format. The scope includes generating a concise summary and highlighting key contract points to reduce the need for full manual document review.

5. Report Generation and User Accessibility

The system generates a structured PDF report containing analysis results, including sector classification, risk level, summary, and simplified terms. The entire application operates on the client side through a web interface, ensuring accessibility without requiring server-side infrastructure.

6. Limitations of the System

The current system is limited to rule-based analysis and predefined keyword detection. It does not incorporate advanced machine learning models, deep semantic understanding, or support for multiple languages. Future enhancements may include AI-based clause classification, predictive risk scoring.

1.4 Literature Review

The increasing complexity of legal contracts has created a demand for automated systems capable of assisting in document analysis and interpretation. Traditional contract review is manual, time-consuming, and requires domain expertise. Researchers have explored the application of Optical Character Recognition (OCR), Natural Language Processing (NLP), and rule-based systems to automate legal document processing and improve efficiency.

Optical Character Recognition technology plays a foundational role in document digitization by converting scanned images into machine-readable text. Tools such as Tesseract OCR have been widely used in document analysis systems to extract textual information from printed and scanned legal documents. OCR enables further computational processing by transforming non-editable documents into analyzable digital text.

Natural Language Processing techniques are commonly applied in legal text analysis for tasks such as tokenization, sentence segmentation, keyword extraction, and document classification. Libraries like NLTK and spaCy provide tools for linguistic processing and information extraction. Advanced research has also explored machine learning and deep learning models for contract classification, clause identification, and risk prediction. However, such approaches often require large annotated datasets and high computational resources.

In addition to machine learning approaches, rule-based and heuristic systems have proven effective for domain-specific applications such as contract analysis. These systems rely on predefined keywords and patterns to identify clauses and detect potential risks. Rule-based models are computationally efficient, easier to implement, and suitable for lightweight client-side applications.

Legal text simplification has also been studied to improve accessibility for non-expert users. Lexical simplification techniques replace complex legal terminology with plain language equivalents, enhancing readability and transparency.

Based on the reviewed literature, integrating OCR with rule-based text processing provides a practical and efficient solution for automated contract analysis. The proposed system builds upon these established concepts by combining OCR-based text extraction, keyword-driven classification, risk detection, clause extraction, and legal term simplification within a unified web-based framework.

CHAPTER 2

METHODOLOGY

2.1 Methodology

The system follows a structured NLP pipeline:

Step 1: Document Upload

User uploads contract document (PDF or text file).

Step 2: Text Extraction

The system extracts raw text from the document.

Step 3: Text Preprocessing

- Convert to lowercase
- Remove punctuation
- Remove stop words
- Tokenize sentences and words
- Perform basic normalization

Step 4: Keyword Extraction

Important legal terms are identified using keyword matching.

Step 5: Clause Identification

The system scans for predefined clause indicators such as:

- Payment terms
- Termination clauses
- Liability clauses
- Confidentiality clauses

Step 6: Risk Detection

Risk-related terms are identified, such as:

- “Penalty”
- “Unlimited liability”
- “Breach”
- “Termination without notice”

Step 7: Risk Scoring

Risk Score = (Number of Risk Keywords × Assigned Weight) / Total Clauses

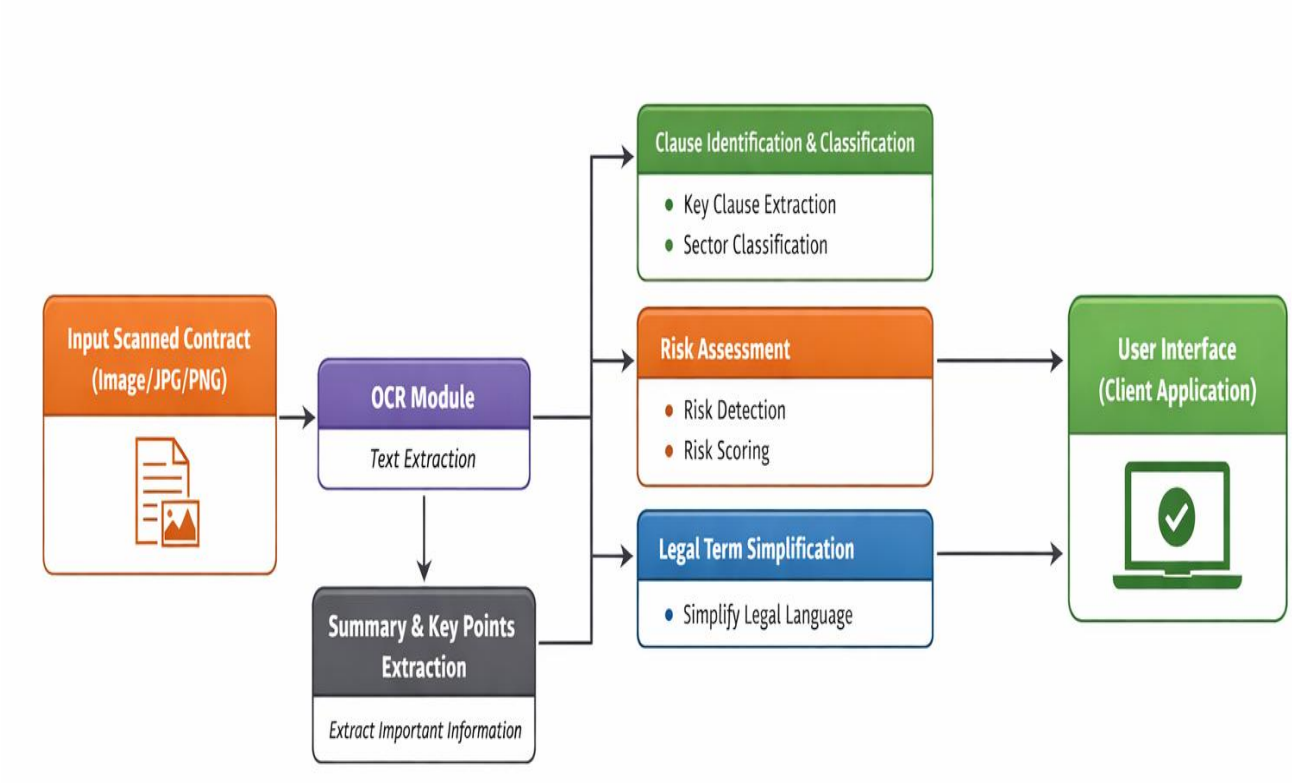
Step 8: Sector Classification
Based on dominant keywords, contract is classified into sectors like:
• Finance
• Employment
• Services
• Real Estate

Step 9: Summary Generation
Important sentences are extracted to form a concise summary.

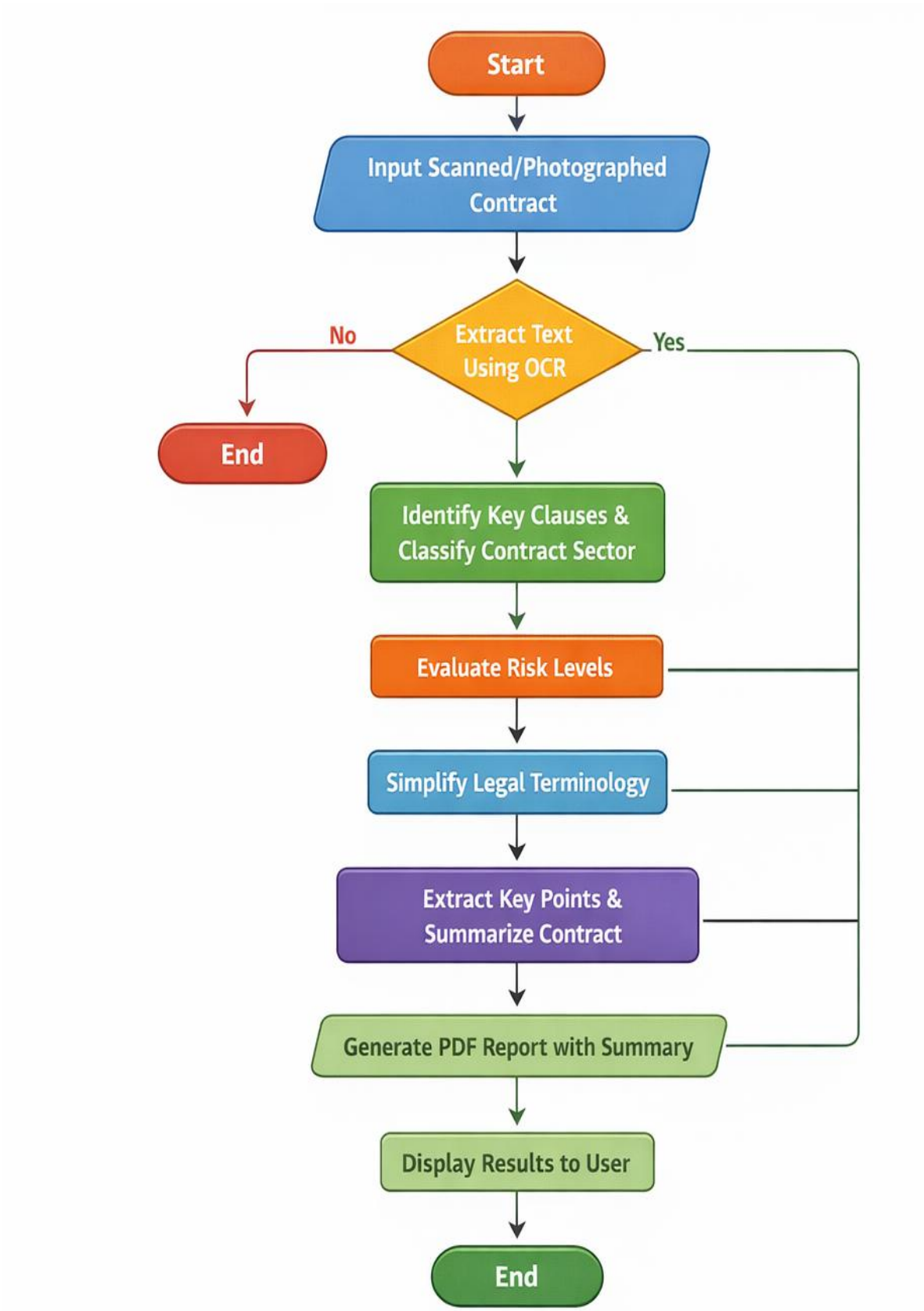
Step 10: Simplification
Complex legal terms are replaced with simpler alternatives.

Step 11: Result Display
All outputs are shown in dashboard format.

2.2 Block Diagram



2.3 Flow Chart



CHAPTER 3

REQUIREMENTS

3.1 Software Requirements

The following software components are required for the development and execution of the OCR - Based Contract Analysis and Simplification System:

1. Development Environment

Visual Studio Code (VS Code) - Used as the primary code editor for writing and managing HTML, CSS, and JavaScript files.

2. Programming Languages

HTML5 - For designing the structure of the web application.

CSS3 - For styling and user interface design.

JavaScript (ES6) - For implementing client-side logic, OCR integration, text processing, and report generation.

3. Libraries and Frameworks

Tesseract.js - For performing Optical Character Recognition (OCR) to extract text from scanned contract images.

Tailwind CSS - For responsive and modern UI styling.

jsPDF - For generating structured PDF reports dynamically.

4. Web Browser

Google Chrome / Microsoft Edge / Mozilla Firefox - Required to run and test the client-side web application.

5. Operating System

Windows 10/11 (or any OS supporting modern web browsers)

6. Internet Connection

Required for loading CDN-based libraries such as Tailwind CSS and Tesseract.js (if not hosted locally).

3.2 Hardware Requirements

The following hardware components are required for the development and execution of the Automated Contract Analysis and Simplification System:

1. Processor

Minimum: Intel Core i3 (or equivalent)

Recommended: Intel Core i5 or higher

2. RAM

Minimum: 4 GB

Recommended: 8 GB or higher (for smooth OCR processing)

3. Storage

Minimum: 10 GB free disk space

Additional space for project files and generated PDF reports

4. Display

Standard monitor with minimum resolution of 1366 × 768

5. Input Devices

Keyboard and Mouse

6. Internet Connectivity

Required for loading external libraries (if using CDN links**)**

CHAPTER 4

IMPLEMENTATION

4.1 Implementation

1. User Interface Development

Developed a responsive web interface using HTML, CSS, and JavaScript. The system includes file upload functionality, loading animation, and a result dashboard to display analysis results. Tailwind CSS was used for styling.

2. OCR-Based Text Extraction

Integrated Tesseract.js to perform Optical Character Recognition (OCR) for extracting text from scanned contract images (JPG/PNG). The extracted text is converted into machine-readable format for further processing.

3. Text Preprocessing

Performed basic preprocessing such as converting text to lowercase and segmenting sentences. This ensures consistent keyword detection and analysis.

4. Clause Detection and Sector Classification

Implemented rule-based keyword matching to identify important clauses and classify contracts into sectors like IT, Finance, Legal, Healthcare, and Construction.

5. Risk Detection Module

Detected risk-related keywords such as penalty, termination, liability, and fine to categorize the contract risk level (High or Medium).

6. Key Point Extraction

Extracted meaningful sentences from the contract text to highlight important clauses and provide a concise overview.

7. Legal Term Simplification

Used a predefined dictionary to replace complex legal terms with simpler, understandable equivalents.

8. Report Generation

Enabled PDF report generation using jsPDF, including sector, risk level, summary, key points, and simplified terms.

4.2 Testing of the Project

4.1.1 Functional Testing

Functional testing was performed to verify that each module of the system operates according to the specified requirements. The file upload feature was tested to ensure that only supported image formats (JPG, JPEG, PNG) are accepted. The OCR module was tested with multiple scanned contract images to confirm accurate text extraction. The clause detection, risk identification, and simplification modules were tested to ensure correct keyword matching and output generation. The PDF report generation feature was also tested to verify proper creation and download of the analysis report.

4.1.2 Module Testing

Each module was tested individually:

User Interface Module - Verified proper file upload, button activation, and display of results.

OCR Module - Tested extraction accuracy with different image qualities.

Risk Detection Module - Verified correct detection of predefined risk keywords.

Simplification Module - Confirmed replacement of complex legal terms with simpler equivalents.

Report Generation Module - Ensured correct formatting and download of PDF reports.

4.1.3 Input and Output Testing

Various contract images with different content lengths and structures were provided as input. The system successfully extracted text, identified key clauses, classified sector type, detected risk level, and generated simplified explanations. Invalid inputs such as unsupported file formats were handled appropriately.

4.1.4 Performance Testing

The system was tested for processing time and responsiveness. OCR processing time varied depending on image size and quality. The application functioned smoothly within standard hardware configurations without significant performance issues.

CHAPTER 5

CODE

5.1 Source Code

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>NLP-Based Automatic Contract Analysis and Simplification System | Upload Contract</title>
  <!-- Tailwind -->
  <script src="https://cdn.tailwindcss.com"></script>
  <!-- Tesseract -->
  <script src="https://cdn.jsdelivr.net/npm/tesseract.js@5/dist/tesseract.min.js"></script>
  <!-- CSS -->
  <link rel="stylesheet" href="style.css">
</head>
<body class="min-h-screen overflow-y-auto">
<div class="page-loader" id="pageLoader">
  <img src="images/logo.png" alt="TextLex Logo">
  <div class="spinner"></div>
  <p>Loading TextLex...</p>
</div>
<div class="intro" id="intro">
  <!-- 🎥 Video -->
  <video
    id="introVideo"
    autoplay
    muted
    playsinline
    preload="auto">
    <source src="videos/intro.mp4" type="video/mp4">
  </video>
  <!-- 🟢 Logo Overlay -->
  <div class="intro-logo">
    <img src="images/logo.png" alt="TextLex Logo">
  </div>
</div>
  <div class="bg-image"></div>
  <div class="blob"></div>
  <div class="blob2"></div>
  <div class="min-h-screen flex justify-center items-start py-16 px-4">
   <div class="glass max-w-xl w-full p-10 rounded-3xl shadow-2xl">
      <h2 class="text-3xl font-extrabold text-indigo-700 mb-2">
        NLP-Based Automatic Contract Analysis and Simplification System
      </h2>
```

```
      <p class="text-gray-500 mb-6">
       simplification of legal documents
      </p>
      <form id="contractForm" class="space-y-6">
        <input
          type="file"
          id="fileInput"
          accept=".jpg,.jpeg,.png"
          class="w-full p-3 border rounded-xl"
        >
        <button
          id="submitBtn"
          disabled
          class="w-full py-3 rounded-xl text-white font-semibold
          bg-indigo-600 opacity-50 cursor-not-allowed"
        >
          Analyze Contract
        </button>    </form>
<div id="progressBox" class="hidden mt-6 analyze-loader">
  <div class="analyze-spinner"></div>
  <p class="analyze-text">
    Analyzing contract, please wait…
  </p>
  <div class="w-full bg-gray-200 rounded-full h-2 overflow-hidden">
    <div
      id="progressBar"
      class="bg-indigo-600 h-2 rounded-full w-0 transition-all duration-300"
    ></div>
  </div>
</div>
  </div>
<audio id="processingSound" loop>
  <source src="sounds/processing.mp3" type="audio/mpeg">
</audio>
  <!-- JS -->
  <script src="script.js"></script>
<script>
  const loader = document.getElementById("pageLoader");
  window.addEventListener("load", () => {
 setTimeout(() => {
      loader.style.opacity = "0";
      loader.style.pointerEvents = "none";
      setTimeout(() => loader.remove(), 600);
    }, 800); // smooth delay
  });
</script>
```

```
<script>
    const intro = document.getElementById("intro");
    const v = document.getElementById("introVideo");
    v.play().catch(() => {});
    setTimeout(() => {
      intro.style.opacity = "0";
      intro.style.pointerEvents = "none";
      setTimeout(() => intro.remove(), 800);
    }, 5000);
  </script>
</body>
</html>
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <title>NLP-Based Automatic Contract Analysis and Simplification System | Analysis Result</title>
  <!-- Tailwind -->
  <script src="https://cdn.tailwindcss.com"></script>
  <!-- jsPDF -->
  <script src="https://cdnjs.cloudflare.com/ajax/libs/jspdf/2.5.1/jspdf.umd.min.js"></script>
  <!-- Custom CSS -->
  <link rel="stylesheet" href="style.css" />
</head>
<body class="min-h-screen overflow-y-auto">
  <!-- 🔥 BACKGROUND IMAGE -->
  <div class="bg-image"></div>
  <!-- Optional blobs (keep if you want) -->
  <div class="blob"></div>
  <div class="blob2"></div>
  <!-- MAIN CONTENT -->
  <div class="min-h-screen flex justify-center items-start py-16 px-4">
    <div class="glass max-w-3xl w-full p-10 rounded-3xl shadow-2xl">
      <h2 class="text-3xl font-extrabold text-indigo-700 mb-6">
        Contract Analysis Result
      </h2>
      <!-- INFO CARDS -->
      <div class="grid grid-cols-2 gap-4 mb-6">
        <div class="card">
          <p class="label">Sector</p>
          <p id="sector" class="value"></p>
        </div>
        <div class="card">
          <p class="label">Risk Level</p>
          <p id="risk" class="value"></p>
```

```
</div>
    </div>
    <!-- SUMMARY -->
    <div class="card mb-6">
      <p class="label">Summary</p>
      <p id="summary" class="text-gray-700"></p>
    </div>
    <!-- SIMPLIFIED TERMS -->
    <div class="card mb-6">
      <p class="label">Simplified Language (Plain English)</p>
      <ul id="simpleTerms" class="list-disc pl-5 space-y-1 text-gray-700"></ul>
    </div>
    <!-- KEYWORD SEARCH -->
    <div class="card mb-6">
      <p class="label">Keyword Search</p>
      <input
        id="searchInput"
        type="text"
        placeholder="Search keyword (e.g. payment)"
        class="w-full mt-2 p-2 border rounded-lg focus:ring-2 focus:ring-indigo-400"
      />
      <p id="searchStatus" class="text-sm text-gray-500 mt-1"></p>
    </div>
    <!-- KEY POINTS -->
    <div class="card">
      <p class="label">Key Points</p>
      <ul id="keyPoints" class="list-disc pl-5 space-y-1 text-gray-700"></ul>
    </div>
    <!-- ACTION BUTTONS -->
    <div class="flex flex-col sm:flex-row gap-4 mt-8">
      <button
        onclick="window.location.href='index.html'"
        class="flex-1 px-6 py-3 bg-indigo-600 text-white rounded-xl"
      >
        Analyze Another Contract
      </button>
      <button
        onclick="downloadPDF()"
        class="flex-1 px-6 py-3 bg-green-600 text-white rounded-xl"
      >
        Download PDF Report
      </button>
    </div>
  </div>
 </div>
<script>
```

```
documentaddEventListener("DOMContentLoaded", () => {
    const stored = sessionStorage.getItem("analysisResult");
    if (!stored) {
      window.location.href = "index.html";
      return; }
    const data = JSON.parse(stored);
    const sectorEl = document.getElementById("sector");
    const riskEl = document.getElementById("risk");
    const summaryEl = document.getElementById("summary");
    const keyPointsUL = document.getElementById("keyPoints");
    const simpleTermsUL = document.getElementById("simpleTerms");
    const searchInput = document.getElementById("searchInput");
    const searchStatus = document.getElementById("searchStatus");
    sectorEl.textContent = data.sector;
    riskEl.textContent = data.risk;
    summaryEl.textContent = data.summary;
    const originalPoints = data.keyPoints || [];
    renderList(keyPointsUL, originalPoints);
    generateSimpleLanguage(originalPoints);
    searchInput.addEventListener("input", () => {
      const q = searchInput.value.toLowerCase().trim();
      if (!q) {
        renderList(keyPointsUL, originalPoints);
        searchStatus.textContent = "";
        return}
      const filtered = originalPoints.filter(p =>
        p.toLowerCase().includes(q)  );
      if (filtered.length === 0) {
        keyPointsUL.innerHTML = "";
        searchStatus.textContent = "No matching clauses found";
      } else {
        renderList(keyPointsUL, filtered);
        searchStatus.textContent = `${filtered.length} match(es) found`;}
    });
    window.downloadPDF = function () {
      const { jsPDF } = window.jspdf;
      const pdf = new jsPDF();
      let y = 15;
      pdf.setFontSize(18);
      pdf.text("NLP-Based Automatic Contract Analysis and Simplification System – Contract Analysis Report",
10, y);
      y += 10;
      pdf.setFontSize(12);
      pdf.text(`Sector: ${data.sector}`, 10, y); y += 7;
```

```
 pdf.text(`Risk Level: ${data.risk}`, 10, y); y += 10;
       pdf.setFontSize(14);
       pdf.text("Summary:", 10, y); y += 7;
       pdf.setFontSize(11);
       pdf.text(pdf.splitTextToSize(data.summary, 180), 10, y);
       y += 15;
       pdf.setFontSize(14);
       pdf.text("Key Points:", 10, y); y += 7;
       pdf.setFontSize(11);
       originalPoints.forEach((p, i) => {
         const lines = pdf.splitTextToSize(`${i + 1}. ${p}`, 180);
         pdf.text(lines, 10, y);
         y += lines.length * 6;
       });
       y += 10;
       pdf.setFontSize(14);
       pdf.text("Simplified Language:", 10, y); y += 7;
       pdf.setFontSize(11);
       Array.from(simpleTermsUL.children).forEach((li, i) => {
         const lines = pdf.splitTextToSize(`${i + 1}. ${li.textContent}`, 180);
         pdf.text(lines, 10, y);
         y += lines.length * 6;
       });
       pdf.save("Project_Pal_Report.pdf");
     };
     function renderList(ul, items) {
       ul.innerHTML = "";
       items.forEach(i => {
         const li = document.createElement("li");
         li.textContent = i;
         ul.appendChild(li); });   }
     function generateSimpleLanguage(points) {
       const dictionary = {
         termination: "Ending the contract",
         liability: "Legal responsibility",
         jurisdiction: "Court that handles disputes",
         indemnity: "Protection against loss",
         penalty: "Fine or punishment",
         confidential: "Must be kept secret",
         obligation: "Duty or responsibility",
         breach: "Breaking the agreement",
         compensation: "Money paid for loss",
         compliance: "Following the rules"  };
       const found = new Set();
       points.forEach(p => {
for (const word in dictionary) {
```

```
 if (p.toLowerCase().includes(word)) {
            found.add(`${word.toUpperCase()} → ${dictionary[word]}`); }
        });
        if (found.size === 0) {
          found.add("No complex legal terms detected");}
        renderList(simpleTermsUL, Array.from(found)); }});
  </script>
</body>
</html>
Script.js
document.addEventListener("DOMContentLoaded", () => {
  const fileInput = document.getElementById("fileInput");
  const submitBtn = document.getElementById("submitBtn");
  const progressBox = document.getElementById("progressBox");
  const progressBar = document.getElementById("progressBar");
  const form = document.getElementById("contractForm");
  if (!fileInput || !form) return;
  fileInput.addEventListener("change", () => {
    submitBtn.disabled = false;
    submitBtn.classList.remove("opacity-50", "cursor-not-allowed");
  });
form.addEventListener("submit", async (e) => {
  e.preventDefault();
  const sound = document.getElementById("processingSound");
  sound.volume = 0.25; //  soft volume
  sound.play().catch(() => {});
  submitBtn.disabled = true;
  submitBtn.textContent = "Analyzing…";
  progressBox.classList.remove("hidden");
  progressBar.style.width = "5%";
  const file = fileInput.files[0];
  if (!file) return;
  const { data } = await Tesseract.recognize(file, "eng", {
    logger: m => {
      if (m.progress) {
        progressBar.style.width = `${Math.floor(m.progress * 100)}%`;
  });
  sound.pause();
  sound.currentTime = 0;
  const result = analyzeText(data.text);
  sessionStorage.setItem("analysisResult", JSON.stringify(result));
  window.location.href = "result.html";
});
const result = analyzeText(data.text);
sessionStorage.setItem("analysisResult", JSON.stringify(result));
    window.location.href = "result.html";
```

```
  function analyzeText(text) {
    const textLower = text.toLowerCase();
    const sectors = {
      IT: ["software", "data", "technology"],
      Finance: ["payment", "invoice", "interest"],
      Legal: ["agreement", "jurisdiction", "law"],
      Healthcare: ["medical", "hospital", "patient"],
      Construction: ["site", "contractor", "material"]  };
    let sector = "General";
    for (const s in sectors) {
      if (sectors[s].some(k => textLower.includes(k))) {
        sector = s;
        break; } }
    const riskWords = ["penalty", "terminate", "liability", "fine"];
    const risk = riskWords.some(w => textLower.includes(w)) ? "High" : "Medium";
    const points = text
      .split(".")
      .map(s => s.trim())
      .filter(s => s.length > 30)
      .slice(0, 8);
    return {
      sector,
      risk,
      summary:
        "This contract defines responsibilities, payment terms, timelines, and legal obligations between the parties.",
      keyPoints: points.length ? points : [
        "Payment terms and deadlines are specified.",
        "Termination clauses are included.",
        "Legal obligations are defined." ] };}
Style.css
.bg-image {
  position: fixed;
  inset: 0;
  background:
    linear-gradient(
      rgba(248, 250, 252, 0.92),
      rgba(248, 250, 252, 0.92)),
    url("https://images.unsplash.com/photo-1614850523459-c2f4c699c52f");
  background-size: cover;
  background-position: center;
  z-index: -2;}
.glass {
  background: rgba(255, 255, 255, 0.78);
  backdrop-filter: blur(26px);
  border-radius: 24px;
  border: 1px solid rgba(255, 255, 255, 0.45);
```

```
 box-shadow:
    0 30px 60px rgba(0, 0, 0, 0.1),
    inset 0 1px 0 rgba(255, 255, 255, 0.6);}
.card {
  background: white;
  padding: 1.25rem;
  border-radius: 16px;
  border: 1px solid #eef2ff;
  box-shadow: 0 10px 25px rgba(0,0,0,0.05);
  transition: transform 0.2s ease, box-shadow 0.2s ease;}
.card:hover {
  transform: translateY(-2px);
  box-shadow: 0 18px 35px rgba(0,0,0,0.08);}
.label {
  font-size: 0.75rem;
  font-weight: 600;
  color: #4f46e5;
  text-transform: uppercase;
  letter-spacing: 0.08em;}
.value {
  font-size: 1.3rem;
  font-weight: 700;
  color: #111827;}
input[type="text"], input[type="file"] {
  background: rgba(249, 250, 251, 0.9);
  border: 1px solid #e5e7eb;
  border-radius: 12px;
  padding: 10px 12px;}
input:focus {
  outline: none;
  border-color: #6366f1;
  box-shadow: 0 0 0 3px rgba(99,102,241,0.2);}
button {
  transition: all 0.25s ease;
button:hover {
  transform: translateY(-1px);}
button:active {
  transform: scale(0.97);}
.bg-image {
  position: fixed;
  inset: 0;
  background-image:
    linear-gradient(
      rgba(0,0,0,0.25),
      rgba(0,0,0,0.25)  ),
    url("https://i.postimg.cc/xT8jR6cF/pexels-rdne-7841478.jpg");
```

```
 background-size: cover;
  background-position: center;
  background-repeat: no-repeat;
  z-index: -10;}
.bg-image {
  filter: blur(3px);
  transform: scale(1.05);}
.bg-image {
  position: fixed;
  inset: 0;
  background-image:    url("https://i.postimg.cc/DyNGdJyb/good-service-cooperation-of-consultation-between-a-male-lawyer-and-business-woman-customer-handshake.jpg");
  background-size: cover;
  background-position: center;
  background-repeat: no-repeat;
  z-index: -10;}
.bg-image::after {
  content: "";
  position: absolute;
  inset: 0;
  background: rgba(0,0,0,0.2);}
@keyframes fadeScale {
  from {
    opacity: 0;
    transform: scale(0.9); }
  to {
    opacity: 1;
    transform: scale(1);  }}
body {
  margin: 0;
  background: black;
  overflow: hidden;}
.bg-video {
  position: fixed;
  inset: 0;
  width: 100%;
  height: 100%;
  object-fit: cover;
  z-index: 1;}
.intro {
  position: fixed;
  inset: 0;
  background: black;
  z-index: 9999;
  transition: opacity 0.8s ease;}
.intro video {
```

```
width: 100%;
  height: 100%;
  object-fit: cover;}
.intro {
  position: fixed;
  inset: 0;
  background: black;
  z-index: 9999;
  transition: opacity 0.8s ease;}
.intro video {
  width: 100%;
  height: 100%;
  object-fit: cover;}
.intro-logo {
  position: absolute;
  inset: 0;
  display: flex;
  align-items: center;
  justify-content: center;
  z-index: 10;
  pointer-events: none;}
.intro-logo img {
  width: 460px;
  max-width: 100%;
  animation: logoFade 1.5s ease;}
.page-loader {
  position: fixed;
  inset: 0;
  background: #0b0f1a;
  z-index: 9000;
  display: flex;
  flex-direction: column;
  align-items: center;
  justify-content: center;
  gap: 20px;
  transition: opacity 0.6s ease;}
.page-loader img {
  width: 120px;
  animation: pulse 1.5s ease-in-out infinite;}
.page-loader p {
  color: #c7d2fe;
  font-size: 0.95rem;
  letter-spacing: 0.08em;}
.spinner {
  width: 38px;
  height: 38px;
```

```
border: 4px solid rgba(255,255,255,0.2);
  border-top: 4px solid #6366f1;
  border-radius: 50%;
  animation: spin 1s linear infinite;}
@keyframes spin {
  to { transform: rotate(360deg); }}
@keyframes pulse {
  0%, 100% { transform: scale(1); opacity: 1; }
  50% { transform: scale(1.08); opacity: 0.7; }}
.analyze-loader {
  display: flex;
  flex-direction: column;
  align-items: center;
  gap: 14px;}
.analyze-spinner {
  width: 42px;
  height: 42px;
  border: 4px solid rgba(99,102,241,0.25);
  border-top: 4px solid #6366f1;
  border-radius: 50%;
  animation: analyzeSpin 1s linear infinite;}
.analyze-text {
  font-size: 0.9rem;
  color: #4f46e5;
  font-weight: 500;
  letter-spacing: 0.03em;}
@keyframes analyzeSpin {
  to {
    transform: rotate(360deg);}}
```

5.2 Screenshots of Application



FIG 5.2.1 : Intro Page



FFIG 5.2.2 : Contract File Upload Page



FIG 5.2.3 : Analyzing Contract File

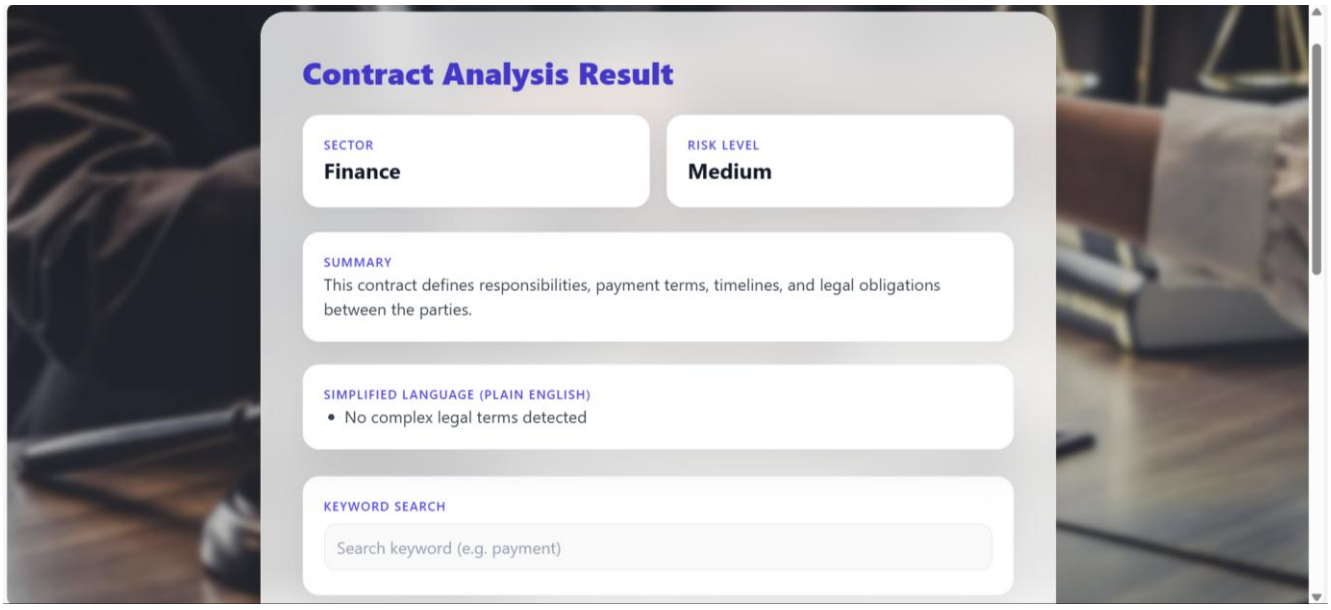


FIG 5.2.4 : Result Page

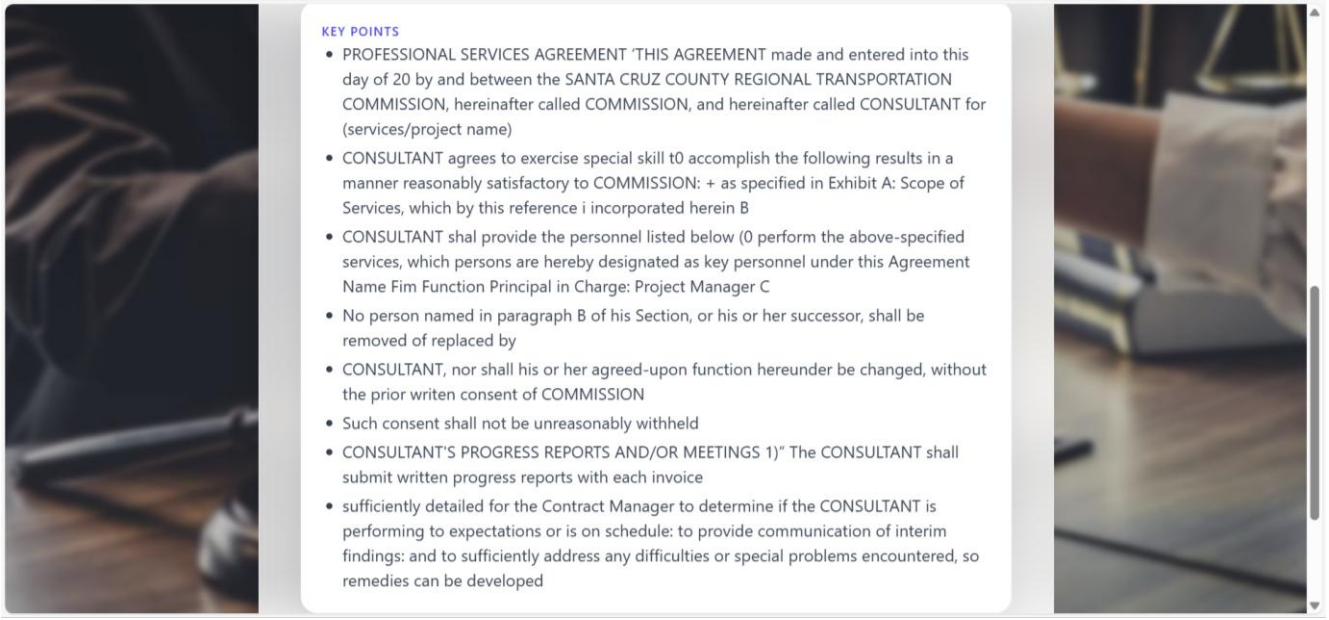


FIG 5.2.5 : Result Page

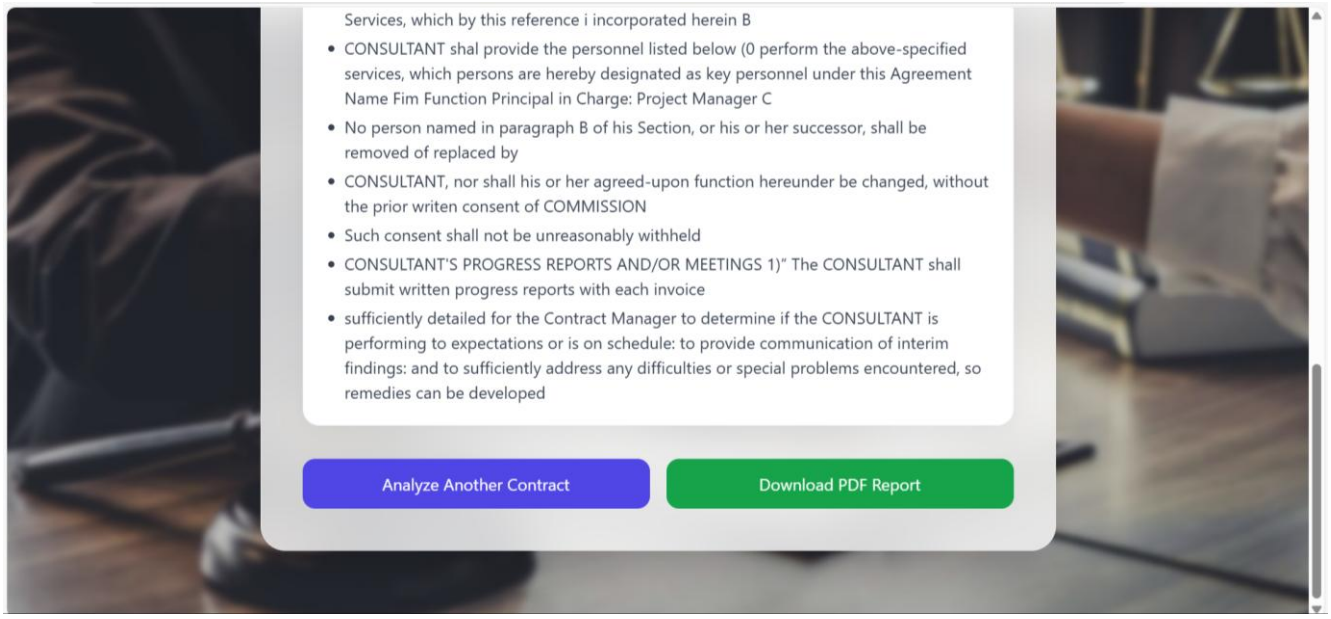


FIG 5.2.6 : Result Page

CHAPTER 6

6.1 Results

6.1.1 Functional Results

The system successfully uploads contract images, extracts text using OCR, and performs contract analysis. Sector classification, risk detection, key point extraction, and simplification are generated correctly.

6.1.2 OCR Results

The OCR module extracts readable text from scanned contract images. Accuracy is good for clear images.

6.1.3 Risk Detection Results

The system detects predefined risk keywords such as penalty, liability, and termination and classifies the contract risk level.

6.1.4 Simplification Results

Complex legal terms are replaced with simpler equivalents using a predefined dictionary.

6.1.5 Report Generation Results

The system generates a structured PDF report containing sector, risk level, summary, and key points.

6.2 Conclusion

The OCR-Based Contract Analysis and Simplification System successfully demonstrates the application of Optical Character Recognition and rule-based text processing for automated legal document analysis. The system converts scanned contract images into machine-readable text and transforms them into structured, simplified, and meaningful outputs.

The system includes:

- Automation of contract text extraction
- Rule-based contract analysis
- Risk keyword detection
- Legal term simplification
- PDF report generation
- User-friendly web interface

The integration of Tesseract.js, JavaScript, and browser-based technologies results in a lightweight, efficient, and accessible contract analysis solution.

Key Contributions

The system provides:

• Automated extraction of text from scanned contracts

• Quick identification of risk-related clauses

• Simplified interpretation of complex legal terms

• Structured summary and key point extraction

• Reduced manual effort in contract review

The platform bridges the gap between complex legal language and user understanding by delivering simplified and organized outputs.

Practical Impact

The system demonstrates how OCR-based automation can:

• Improve understanding of legal documents

• Reduce manual contract review effort

• Enhance transparency in agreements

• Support faster contract evaluation

The project proves that automated contract analysis can be implemented effectively using lightweight web technologies.

Limitations

Despite successful implementation, certain limitations exist:

• Accuracy depends on quality of scanned images

• Rule-based analysis may not capture deep semantic meaning

• Limited to predefined keywords and patterns

These limitations provide opportunities for further improvement.

Future Enhancements

The system can be extended with:

• Machine learning-based clause classification

• Advanced risk scoring mechanism

• Support for PDF and DOCX formats

• Multilingual contract analysis

• Integration with cloud storage

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